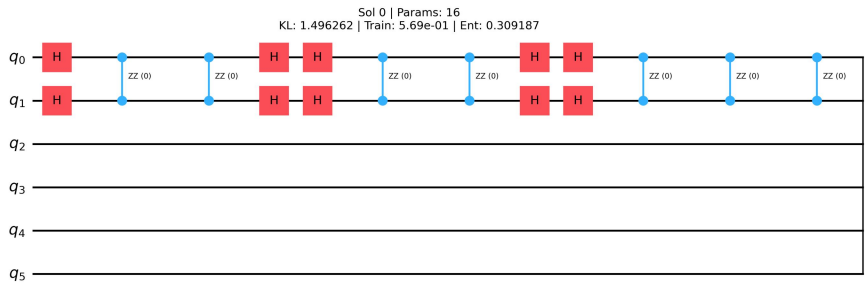
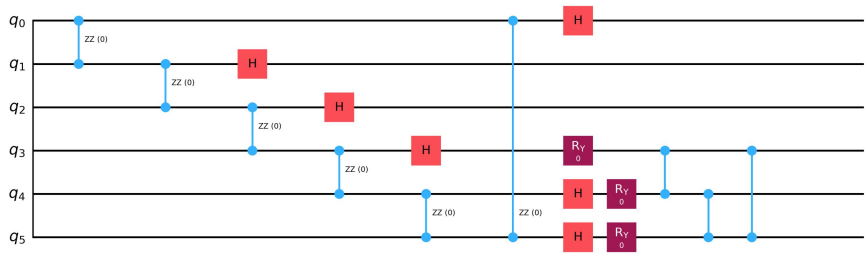
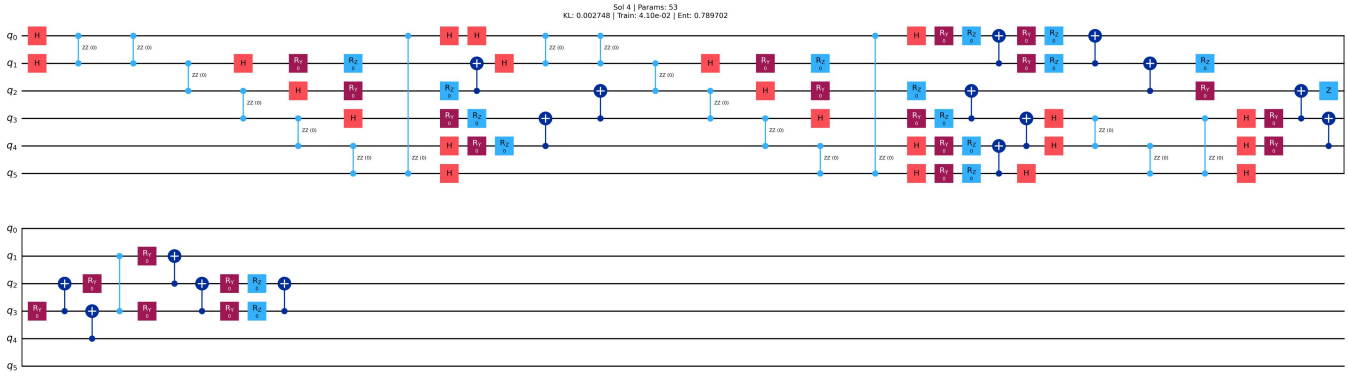
## A. *EQAS Search Results (6 Qubits)*

Fifteen Pareto-optimal 6-qubit VQCs were identified using EQAS, as depicted in Fig. 6 in the main text. VQC 13 was ultimately chosen for the VQFE-DR implementation. For comparison, Fig. S3 presents the circuit structures of this selected VQC, alongside VQC 0, which demonstrated the highest trainability, and VQC 4, which exhibited the greatest expressibility.



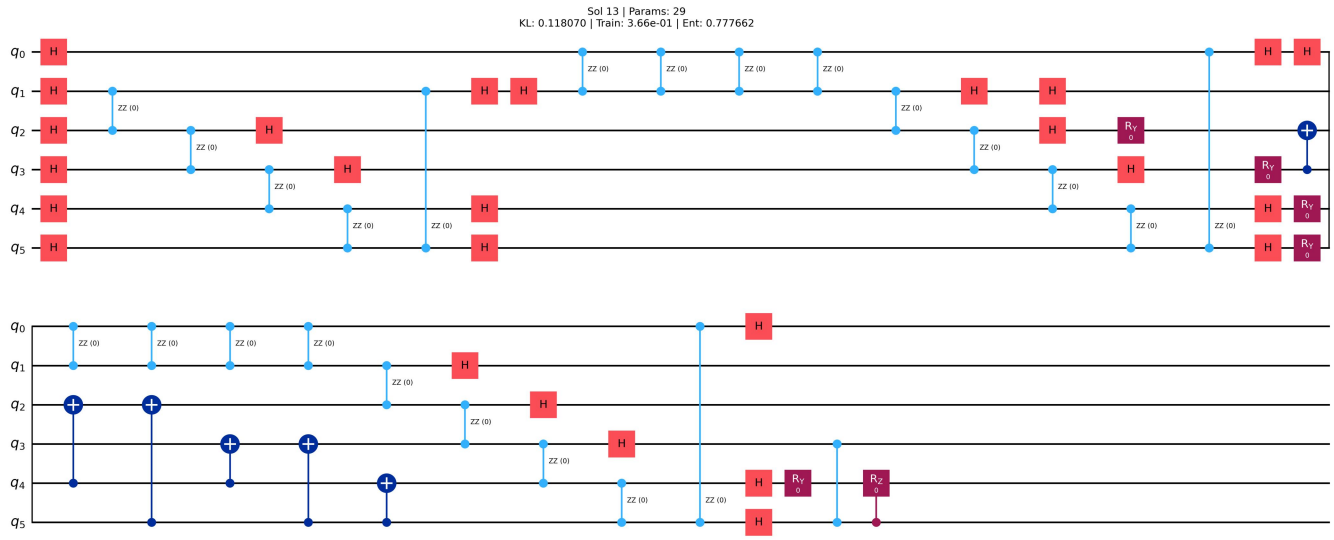
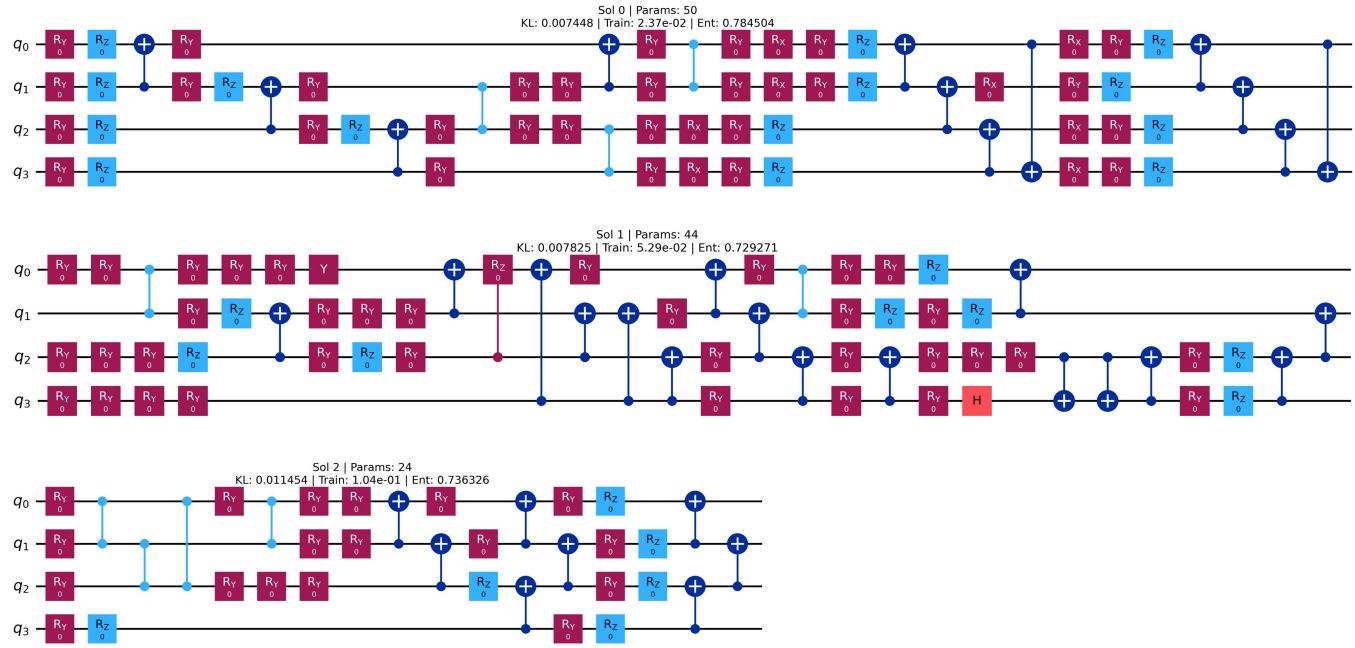


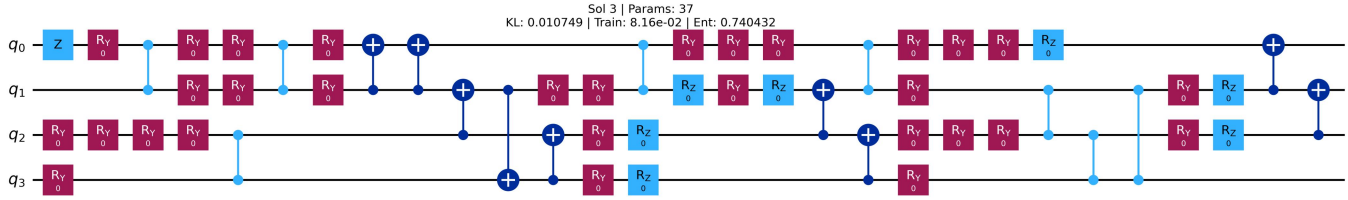

**Fig. S1.** Illustration of key EQAS-identified 6-qubit circuit designs. The implemented VQC 13 is compared with VQC 0 (optimal trainability) and VQC 4 (optimal expressibility).

## B. *EQAS Search Results (4 Qubits)*

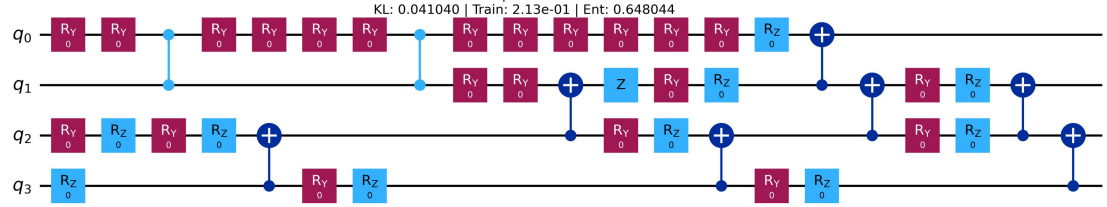
For the 4-qubit VQC, eighteen quantum circuits were initially identified via EQAS. Subsequently, QSeqDAS was employed to perform a fine-grained search on these candidates, demonstrating the heterogeneous VQC requirements across different QLSTM gates. The circuit structures presented below correspond to the EQAS-identified candidates, with sequence numbers consistent with Fig. 7 in the main text.



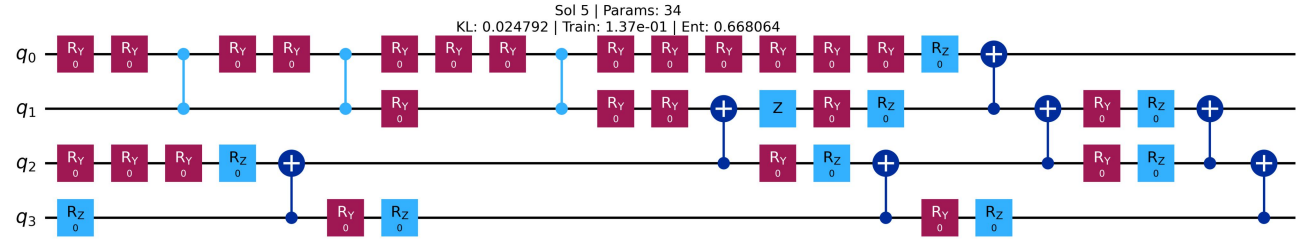
Sol 3 | Params: 37
KL: 0.010749 | Train: 8.16e-02 | Ent: 0.740432



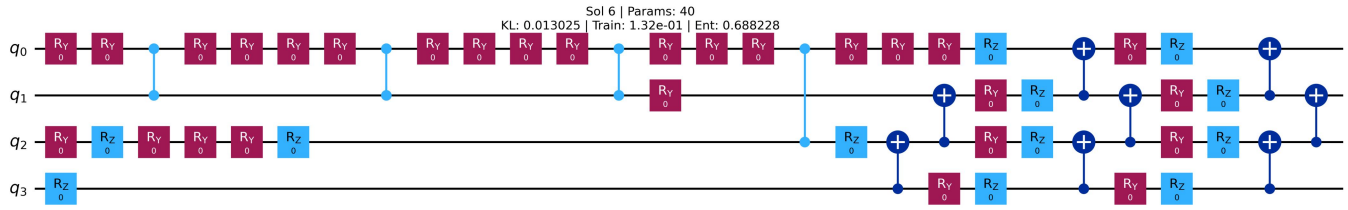
Sol 4 | Params: 32
KL: 0.041040 | Train: 2.13e-01 | Ent: 0.648044



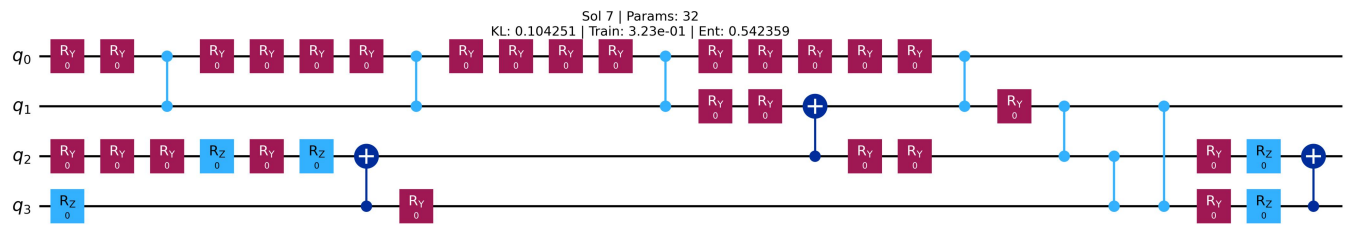
Sol 5 | Params: 34
KL: 0.024792 | Train: 1.37e-01 | Ent: 0.668064



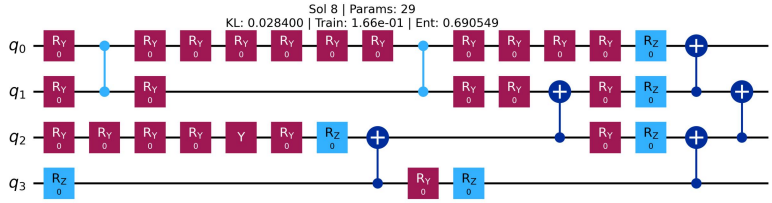
Sol 6 | Params: 40
KL: 0.013025 | Train: 1.32e-01 | Ent: 0.688228



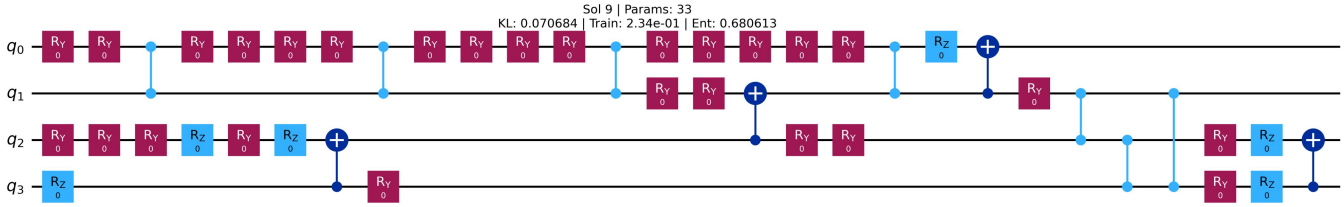
Sol 7 | Params: 32
KL: 0.104251 | Train: 3.23e-01 | Ent: 0.542359



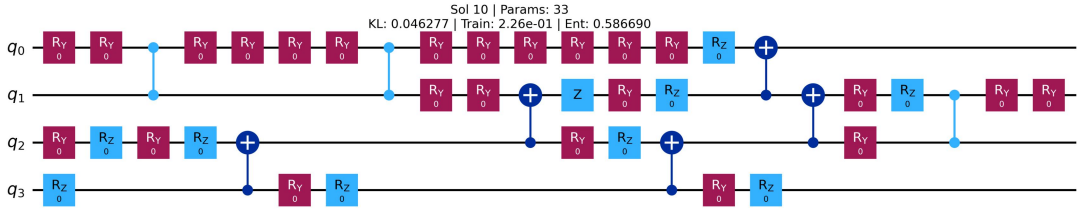
Sol 8 | Params: 29
KL: 0.028400 | Train: 1.66e-01 | Ent: 0.690549



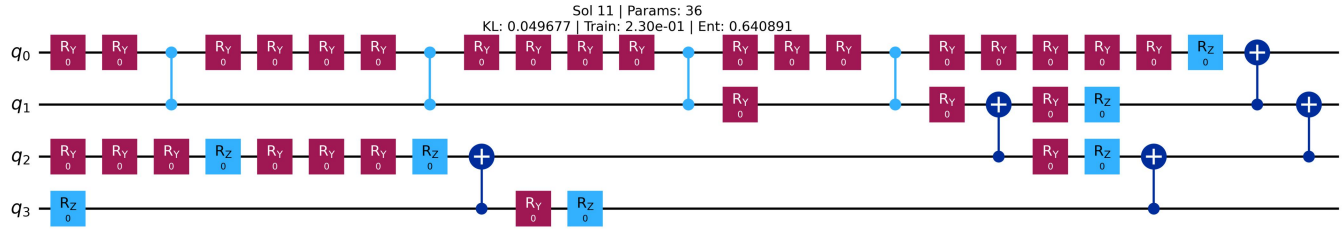
Sol 9 | Params: 33
KL: 0.070684 | Train: 2.34e-01 | Ent: 0.680613



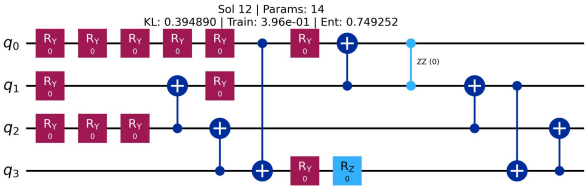
Sol 10 | Params: 33
KL: 0.046277 | Train: 2.26e-01 | Ent: 0.586690



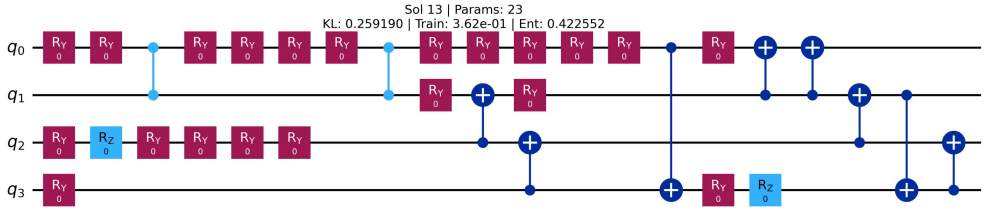
Sol 11 | Params: 36
KL: 0.049677 | Train: 2.30e-01 | Ent: 0.640891



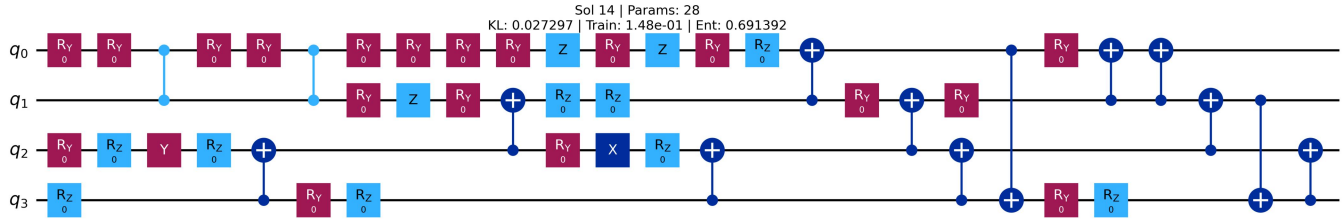
Sol 12 | Params: 14
KL: 0.394890 | Train: 3.96e-01 | Ent: 0.749252




Sol 13 | Params: 23
KL: 0.259190 | Train: 3.62e-01 | Ent: 0.422552



Sol 14 | Params: 28
KL: 0.027297 | Train: 1.48e-01 | Ent: 0.691392



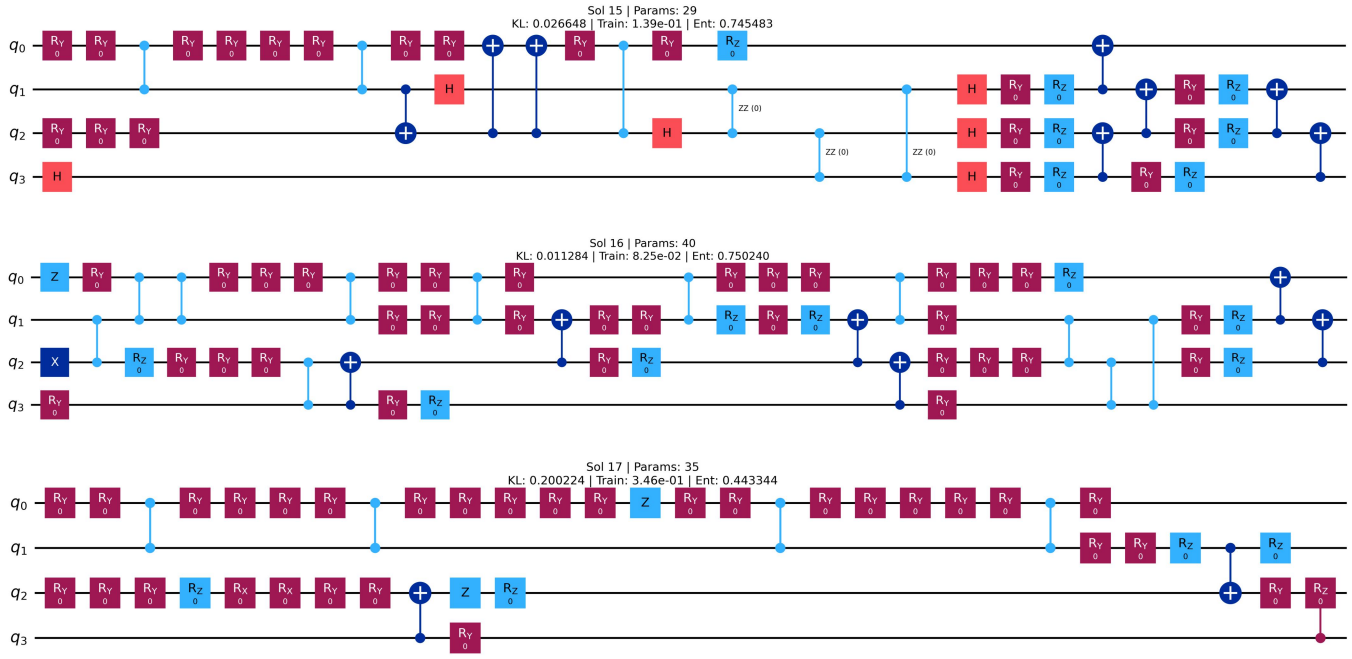



**Fig. S2.** The eighteen 4-qubit VQCs identified via EQAS, serving as candidates for the subsequent QSeqDAS fine-grained search targeting QLSTM gates.